

Database Management Systems

CPS 3740

Lecture 03

Computer Science, Kean University

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Outline

- Continue last lecture
- Chapter 2, 29
 - 3-tier architecture database system
 - Database model and SQL
- Linux
- HTML, PHP
 - Build a web page on the application server
 - Personal website
- Project

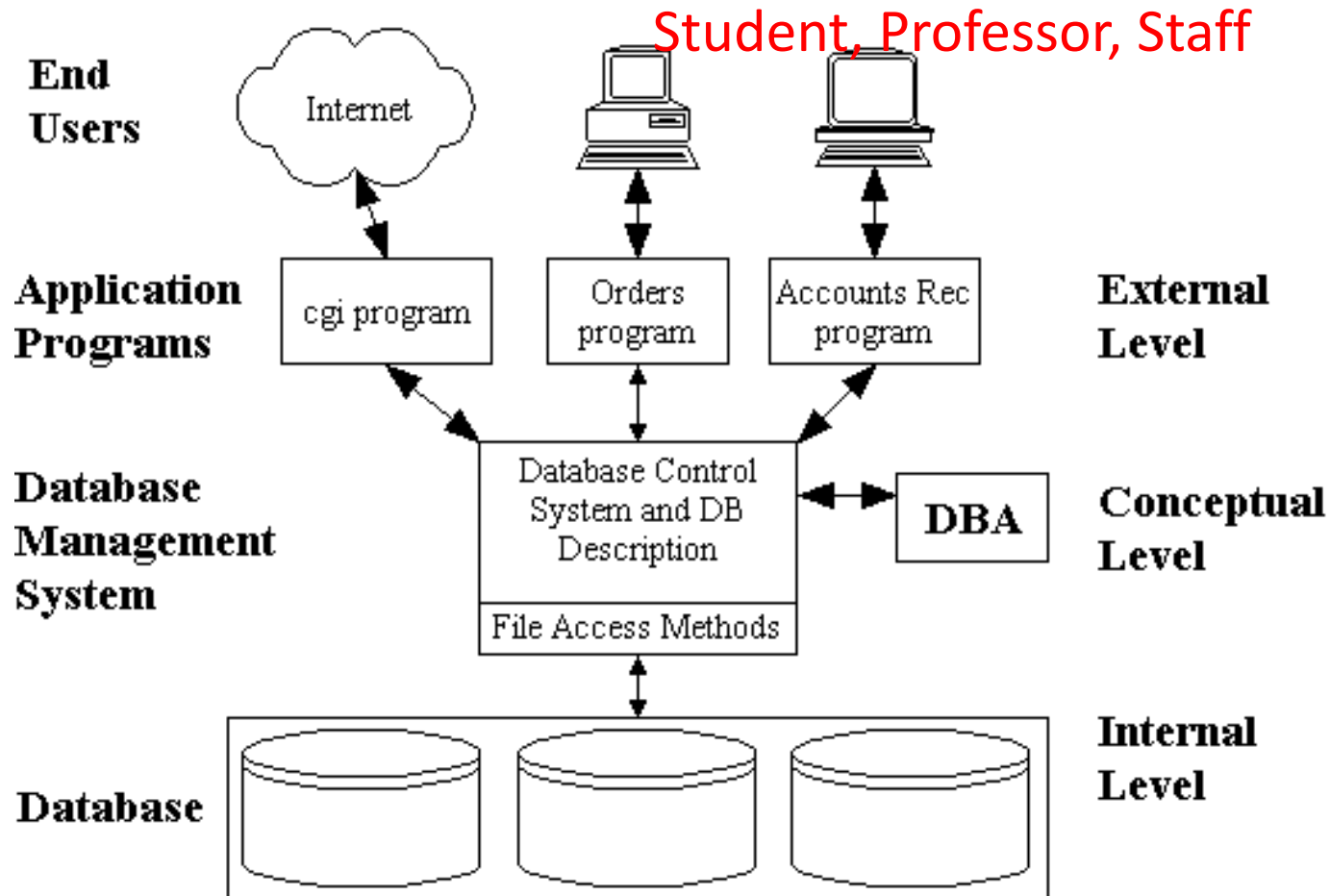
Chapter 2

Database Environment
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The Three Schema Architecture

Proposed by ANSI/SPARC in 1975

- fits most DB models: e.g. Relational, Network, Hierarchical
- divides the system into three levels, external, conceptual and internal:



ANSI-SPARC Three-Level Architecture

- Internal Level
 - Physical representation of the database on the computer.
 - Describes how the data is stored in the database.
- External Level
 - Users' view of the database.
 - Describes that part of database that is relevant to a particular user.
- Conceptual Level
 - Community view of the database.
 - Describes what data is stored in database and relationships among the data.

Objectives of Three-Level Architecture

- All users should be able to access same data.
- A user's view is immune to changes made in other views.
- Users should not need to know physical database storage details.
- DBA should be able to change database storage structures without affecting the users' views.
- Internal structure of database should be unaffected by changes to physical aspects of storage.
- DBA should be able to change conceptual structure of database without affecting all users.

Differences between Three Levels of ANSI-SPARC Architecture

External view 1

sNo	fName	lName	age	salary
-----	-------	-------	-----	--------

External view 2

staffNo	lName	branchNo
---------	-------	----------

Conceptual level

staffNo	fName	lName	DOB	salary	branchNo
---------	-------	-------	-----	--------	----------

Internal level

```
struct STAFF {  
    int staffNo;  
    int branchNo;  
    char fName [15];  
    char lName [15];  
    struct date dateOfBirth;  
    float salary;  
    struct STAFF *next;  
};  
index staffNo; index branchNo;
```

/* pointer to next Staff record */
/* define indexes for staff */

Logical Data Independence

- **Logical** Data Independence
 - Changes to the conceptual schema should not affect the external schema.
 - Conceptual schema changes (e.g. adding/removing tables) should not require changes to external schema or rewrites of application programs.

Physical Data Independence

- **Physical** Data Independence
 - Refers to immunity of conceptual schema to changes in the internal schema.
 - Internal schema changes (e.g. using different file organizations, storage structures/devices) should not require change to conceptual schemas(e.g. tables) or external schemas(e.g. application programs).

Conceptual Modeling

- Conceptual schema is the core of a system supporting all user views.
- Should be complete and accurate representation of an organization's data requirements.
- Conceptual modeling is process of developing a model of information use that is independent of implementation details. (e.g. ER diagram)
- Result is a conceptual data model.

Data Model

Integrated collection of concepts for **describing data, relationships between data, and constraints on the data** in an organization.

- Data Model comprises:
 - a structural part;
 - a manipulative part;
 - possibly a set of **integrity** rules.

Data Model

- Purpose
 - To represent data in an understandable way.
- Categories of data models include:
 - Object-based
 - Record-based
 - Physical.
- Object-Based Data Models
 - Entity-Relationship
 - Semantic
 - Functional
 - Object-Oriented.
- Record-Based Data Models
 - **Relational Data Model**
 - Network Data Model
 - Hierarchical Data Model.
- Physical Data Models

Relational Data Model

Branch

branchNo	street	city	postCode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

- Data and relationships are represented as tables, each has a number of columns and unique name.
- A relation is a table

Staff

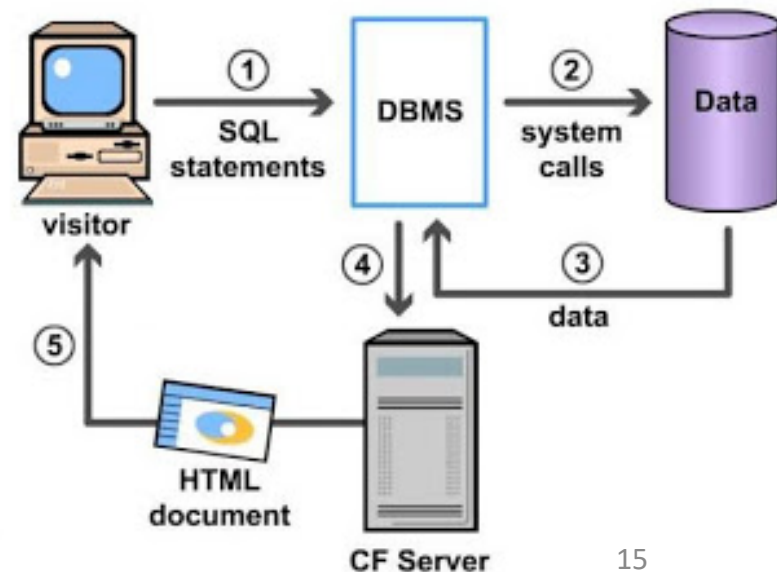
staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

Chapter 29

Web Technology and DBMSs
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Database Management System (DBMS)

- DBMS: Oracle, MySQL, MS SQL Server
- A DBMS can handle many database instances
- DBMS can serve many users (programs, people)
- DBMS functions through SQL
 - Manage users (create, delete, ...)
 - User privilege
 - Manage tables (keys, foreign keys)
 - Manage spaces
 - search, replace, SUM, AVG, ...
 - Backup, restore



How to Display Data?

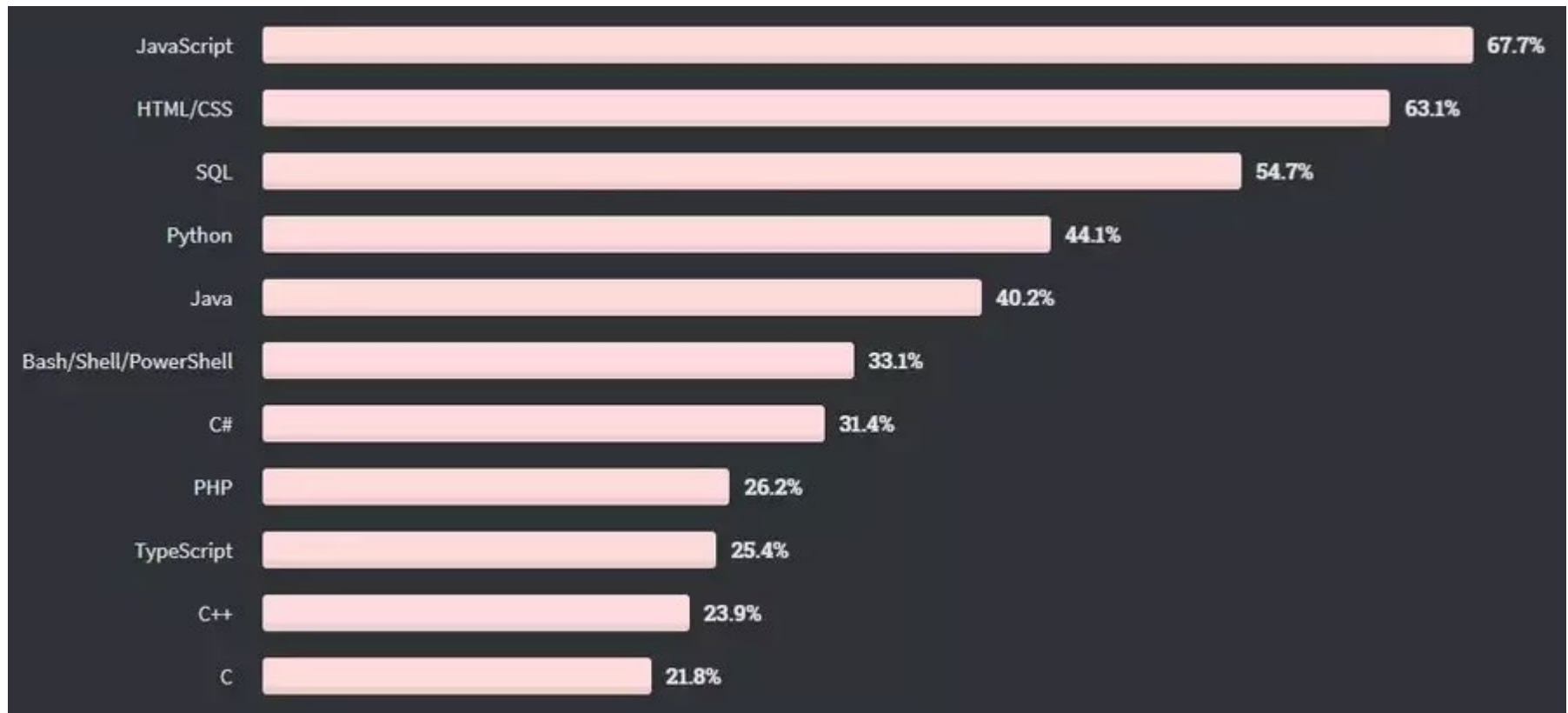
- Data is stored in database.
- Human cannot visualize data without an interface (text or GUI).
- A web application or web app is any software that runs in a web browser
 - It is created in a browser-supported programming language (such as the combination of JavaScript, HTML and CSS) and relies on a web browser to render the application
- Web application (browser based GUI) is much easier to develop than standalone application.
 - That is why our projects are web based.

Web Programming

- Front-end at Browser
 - Graphic User Interface
 - Java, Javascript, HTML, CSS, Flash
- Back-end at Web server
 - Process, access data from database servers.
- Common Gateway Interface (CGI) :
 - Programs implemented on a Web server, provides an interface between the Web server and programs that generate dynamic content on Web pages and Web applications.
 - Usually written in a scripting language, but can be written in any programming language.
 - PHP will be used in our project.

Project – SQL, PHP, HTML, Linux

Most Popular Programming Languages



<https://insights.stackoverflow.com/survey/2020>

PHP, web, database and Linux

- Lab exercise

29.2 Web Pages - HTML

- HyperText Markup Language, commonly referred to as HTML, is the standard markup language used to create web pages. Along with CSS, and JavaScript, HTML is a cornerstone technology, used by most websites to create visually engaging webpages, user interfaces for web applications, and user interfaces for many mobile applications.
- A front-end (browser) language
- Static web pages
- Dynamic web pages
- Interact with database server?



HTML TAG & Attributes

- `<TAG Attributes> ...</TAG>`
- Attributes:
 - `name='xxx'`
 - `value='xxx'`
 - `type='xxx'`
 - `size=xxx`
 - ...
- Single quote ' and double quote "
- Can be either lowercase or uppercase

More about HTML

- HTML Tutorial
 - <http://www.w3schools.com/html/>
- Tag `<...>` , **NOT** case sensitive
- Start with tag `<HTML>`, end with tag `</HTML>`
- File extension: `*.html` or `*.htm`
- Need **read** permission for others, `chmod 704 *.html`
- **User can view HTML FORM source codes.**
- HTML links: `<a href="url">link text</a>`

HyperText Markup Language (HTML)

- Tags & attributes
 - `<TABLE> </TABLE>`
 - `<TR>`, `<TD>`
 - `<FORM> </FORM>`
 - `<SELECT> </SELECT>`
 - `<OPTION> </OPTION>`
 - `<INPUT>`
 - `<BR>`
 - `<UL> </UL>`
 - `<LI>`
 - `<FONT> </FONT>`
 - `<A HREF> </A>`

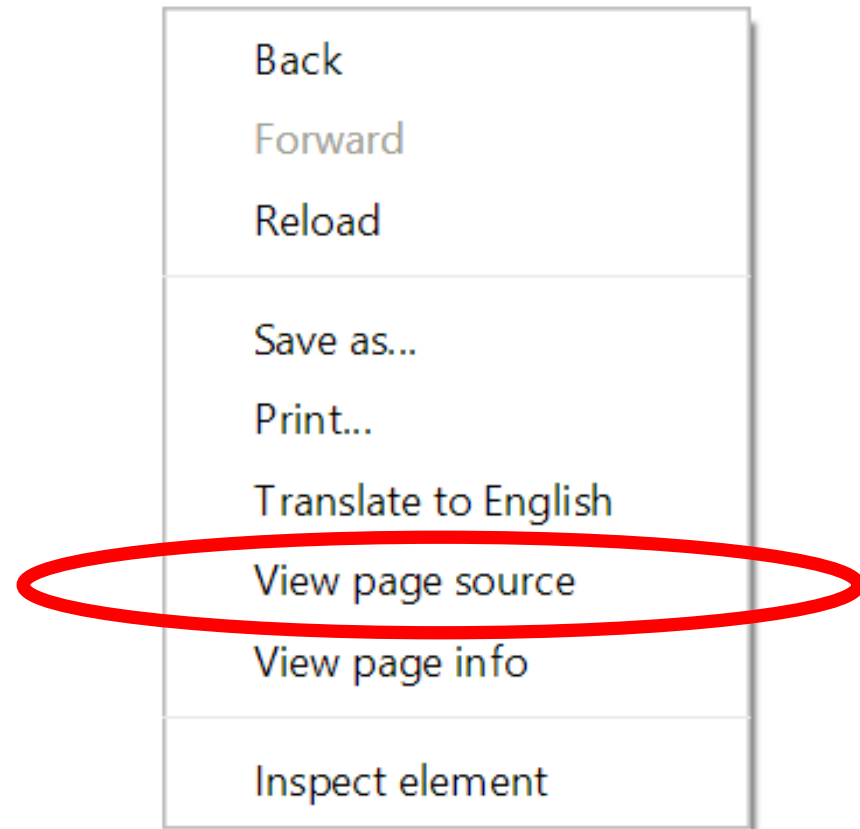
Project demo page

- Demo site
 - <http://vader.kean.edu/CPS3740>
- Use **Sublime** to edit the following code and name the file as **index.html**

```
<HTML>
<h3>Welcome to your_name CPS 3740 Project</h3>
<OL>
<LI><a href="project.html" target="_blank">Project</a>
</OL>
</HTML>
```


View Page Source

- Click right button on browser to view HTML source code
- PHP source code cannot be seen if you view page source on the browser
 - Only HTML codes are available
 - PHP output HTML codes



File Permissions

- Each file/folder has 3 categories:
 - owner, group, others
- Each category has 3 permission modes:
 - Read (r), write (w), execute (x)
 - 1 means on, 0 means off
- `chmod -R` (recursive)
\$ `chmod -R 705 public_html`
- What is the permission mode for 705
 - Owner can read, write and execute, group cannot do anything, and others can read and execute only.
- **Important note: Always ensure the owner's permission is set to 7 (`chmod 7xx <dir/file>`)**

List Files on Linux System

- Under Linux prompt \$, using **ls** command with options **-la** as show below:

```
[www@linux]$ ls -la
```

```
total 16
```

drwxrwxr-x	3	www	www	4096	Oct	1	16:54	.
drwxrwx---	27	www	apache	4096	Oct	1	10:22	..
drwxrwxr-x	2	www	www	4096	Oct	1	16:54	p1
-rwx---r-x	1	www	www	15	Oct	1	16:53	test.php

Parent folder

Current folder

Permission
mode

Owner

Group

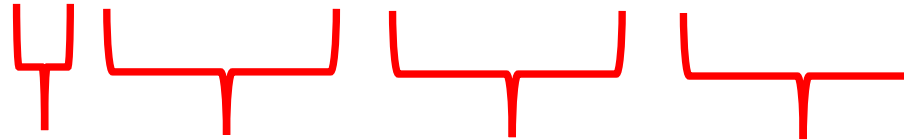
Created
Date & time

File or
folder
name

File Permission Mode

drwxrwxr-x

-rwx---r-x



File type

Owner

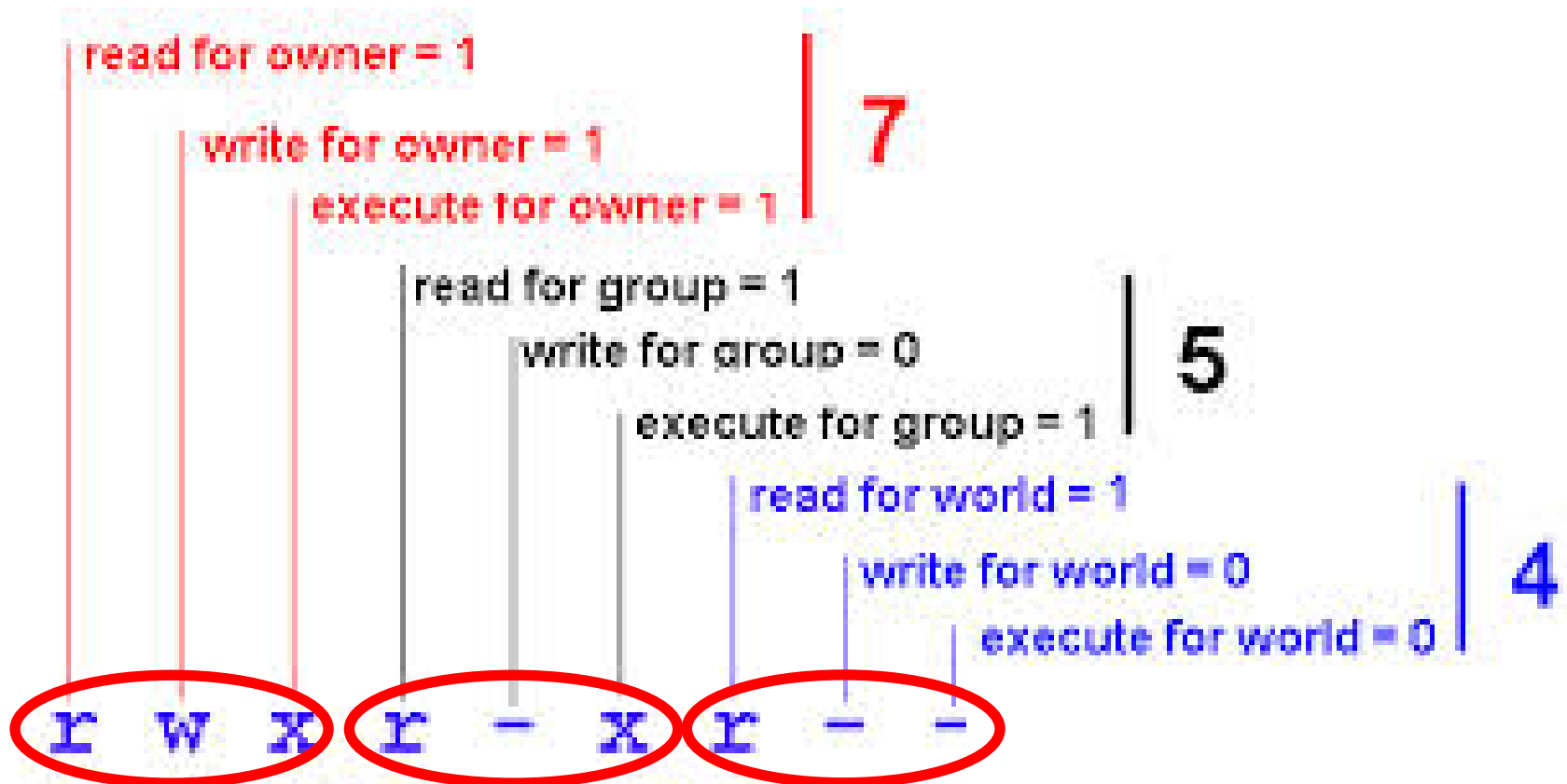
Group

Others

- Home directory, public_html, other sub-folders: 705
- \$ **chmod 705 my_file_or_folder**
- *.html, *.php: 704
- Other folders and files: 700
- Also need to change your home directory: \$ **chmod 705 ~**

Example of Permission Mode

\$chmod 754 my_file



Class exercises - Create Project Home Page

- Follow day 0 slide to login to the webserver and change permission mode.
- File name and folder name: **case sensitive**
 - No space in file or folder names.
- Use sublime text editor to create index.html
 - On Windows or Mac
- Transfer the file index.html to the server
 - **Put all project files under folder ~public_html/CPS3740**
 - You must have **index.html** under CPS3740
- Permission mode for all PHP, html file should be 705

Class exercises - index.html

- Web server will look for **index.html** as the **default html file** under **public_html** and **sub folders**.
- File name is case sensitive in Linux system

<HTML>

<TITLE>xxxxx home page</TITLE>

Welcome to visit xxxxx home page.

</HTML>

- Change permission mode to 705
 - Your home directory
 - Folder: public_html
 - Files: index.html

Working on Linux

- Commands are case sensitive
 - All Unix commands are in lower case
 - Option with -
 - Commands: mv, mkdir, ls, pwd, more, chmod, cd
- File system
 - Tree structure
 - Command: df -h
 - Parent folder: ..
 - Current folder: .
 - Permission mode: Owner, group, other
 - Command: id
 - Content: more, cat, less

Static and Dynamic Web Pages

- HTML document stored in file is static Web page.
- Content of dynamic Web page is generated each time it is accessed.
- Dynamic Web pages:
 - respond to user input from browser;
 - be customized by and for each user.
 - Might have different contents without changing the source codes.
- Requires hypertext to be generated by servers.
 - Data is usually from the database server.
- Need **scripts** (**programs**) that perform conversions from different data formats into HTML '*on-the-fly*'.
 - Server side programming

Interpreting/Compiling Source Code

- Translating **source code** to **machine code** can be done using another programming tool called an *interpreter* or a *compiler*.
- Interpreter
 - Reads **one statement** from the **source code**, translates it to the machine code or virtual machine code, and then **executes it right away**
 - Languages: PHP, perl, shell, Python, Javascript, HTML, ...
- Compiler, two steps
 - Languages: Java, C, C++, ...
 - Compiling source code: **\$javac Test.java**
 - Running a program: **\$java Test**

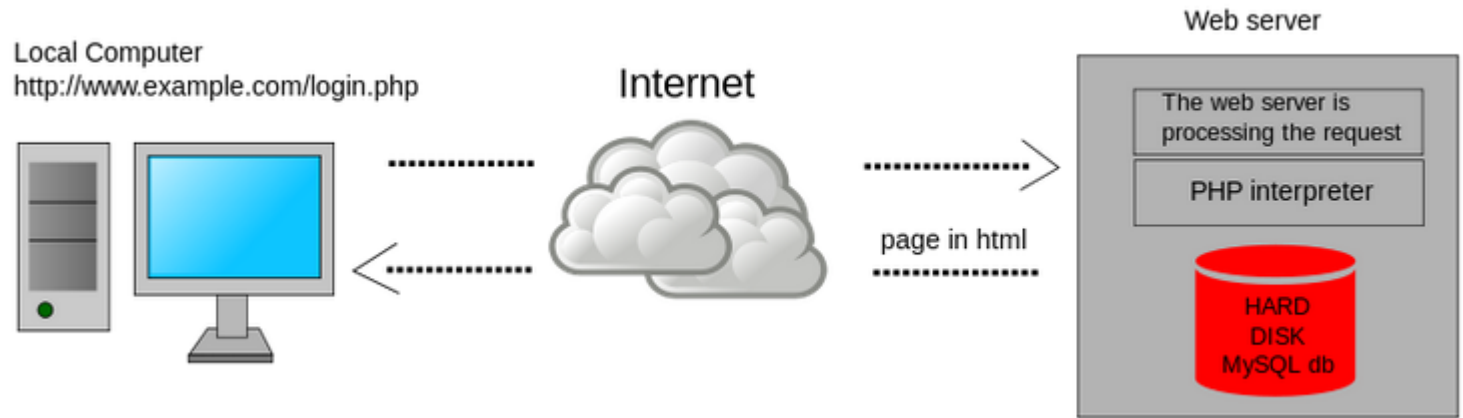
29.3 Script Languages

- Require an interpreter to translate source codes (text) into machine codes (binary)

`$php test.php`

- Browser
 - `<tag> .... </tag>`
- Script language are usually machine independent
 - The interpreter is machine depended.
- Start and stop interpreting the code
 - Start PHP: `<?php`
 - End PHP: `?>`
 - Start html: `<HTML>`
 - End html: `</HTML>`
 - `<font color=red size=5> ... </font>`

PHP



- A popular **server-side scripting** language designed for web development but also used as a general-purpose programming language.
- PHP code can be simply mixed with HTML code, or it can be used in combination with various templating engines and web frameworks.
- Most of new web servers support PHP.
- **This class will only cover basic PHP.**

Basic PHP

- For web, PHP file must be named as *.php
- “;” is the statement terminator.
- Not require to declare a variable.
- Allow to have multiple <?php ?> <?php ?>
- Allow to mix HTML (included Javascript) and PHP codes together.

<HTML>

<?php \$name= "Austin";

\$x=1; \$x="....";

?>

 Welcome <?php echo "\$name". \$x ; ?>

</HTML>

- Print HTML in PHP

echo "
Welcome.\n ";

Notes of PHP

- How do you know php runnable by web server?
 - Browser will show nothing, if there is syntax error.
 - Check if the program has syntax error:
\$ php test.php

PHP & HTML Example

- PHP file name must be *.php

- Live demo

```
<HTML>
```

```
<h3>This is a test</h3>
```

```
<?php
```

```
    echo "hello, this is xxxx!";
```

```
    $a=3;
```

```
    echo "\n<br>$a";
```

```
?>
```

```
</HTML>
```

- Change “xxxx” to your name.
- Remove “;” and see what happens on the browser.

Class exercises – PHP programming

- Assign your first name to a PHP variable \$fname
- Assign your last name to a PHP variable \$lname
- Assign a number 100 to a PHP variable \$a
- Assign a number 150 to a PHP variable \$b
- Print “Kean University”
- Print your name with \$fname and \$lname
- Print the sum of \$a and \$b.

Debug PHP programs

- If the web page not displayed:
 - Syntax error or permission denied
- If you see the following message
 - Warning: Unknown: failed to open stream: Permission denied in Unknown on line 0 Fatal error:
 - Need to change permission mode to 705
 1. Folder /home/student/xxxx/
 2. Folder /home/student/xxxx/public_html
 3. The PHP file
 - Parse error: syntax error, unexpected T_STRING in /home/.... /test.php on line nn
 - Check the line nn and the line above nn

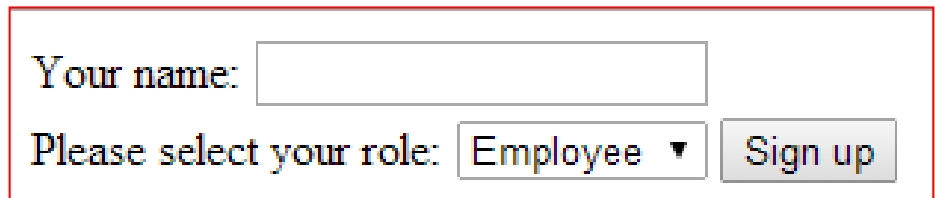
Error or warning messages

- Apache web server will display the warning or error messages of your PHP program on the browser. For example:
 - Warnings: variable, index, array, cookie, sessions are not defined before using them
 - Syntax errors
 - The message will tell you the line#.
- **Fix all the warning and error messages.**
 - You don't want to show these message to the future employer
 - You don't want the hacker to see the details.

Long text strings

- Between opening <<<_HTML_ and closing _HTML_;

```
<?php
print<<<_HTML_
<form method="POST" action="$ _SERVER[PHP_SELF]">
Your name: <input type="text" name="user">
<br>Please select your role:
<select name="role">
  <option value="E">Employee</option>
  <option value="M">Manager</option>
<br>
<input type="submit" value="Sign up">
</form>
_HTML_
?>
```



Your name:

Please select your role: Employee ▼

Single Quote and Double Quote

- Print a HTML link
- Print php variable value
- Print php variable name

```
<?php
```

```
$name="Austin";
```

```
echo "<br><a href='http://www.cnn.com'>CNN</a>\n";
```

```
echo "<br>$name\n";
```

```
echo "<br>'$name'\n";
```

```
echo '<br>$name\n';
```

```
?>
```

CNN
Austin
'Austin'
\$name\n

```
<HTML>
<br><a href='http://www.cnn.com'>CNN</a>
<br>Austin
<br>'Austin'
<br>$name\n
</HTML>
```

Projects Overview

- This project involved the design and implementation of an online information system utilizing MySQL, HTML, and PHP.
- 3-tiers Client-Server
 - Database server(imc.kean.edu)
 - Web application server(obi.kean.edu)
 - User browser

Project Development

- Build a database web application
 - Front-end web pages: HTML
 - Backend: PHP & MySQL
- Build CGI FORM functions
 - Mainly using PHP, PHP generates HTML
- Build PHP programs to access MySQL database
 - Call special DB library functions in specific order
 - PHP programs retrieve data and display in HTML
 - PHP programs to add data in HTML
- Create your own table in the MySQL database, so that your web program can insert data into it.

Review Project

- Demo page <http://vader.kean.edu/CPS3740>
- Computer language: HTML, PHP, SQL
- Write a nice description under the project home page
- Put the project on your resume
 - Short description – A web database application that
 - List the project URL
 - List the skills: HTML, PHP, PHP, MySQL, Linux

HTML Tags in Projects

<http://www.w3schools.com/tags/>

- `<A href="http://yoda.kean.edu/~samurai"> Samurai home</A>`
- defines a hyperlink, link from one page to another
- `<BR>` - Defines a single line break
- `<SELECT></SELECT>` - Defines a drop-down list
- `<OPTION></OPTION>` - Defines an option in a drop-down list
- `<FORM></FORM>` - create an HTML form for user input
- `<INPUT></INPUT>` - specifies an input field where the user can enter data, used within a `<form>`
- `<TABLE></TABLE>` - defines an HTML table.
- `<TITLE></TITLE>`
-

Home Exercise – Website

- Check your grade at class website
- Create your website on the application server
- Login to the server
- Create folder **public_html**
- Use notepad to edit a file index.html
- Use Filezilla to transfer the file to under public_html
- **Set permission 705 for folders and files**

Exercise - Linux commands

- Review the Linux command paper
- File system
 - List files with all details
 - Create folder
 - . and ..
 - Change file/folder name
 - Move a file/folder
 - Delete a file/folder
 - Change file/folder permission
 - Show current full path
 - Change folder location

Other useful commands

- Current time: `date`
- Manual: `man`
- Check system uptime: `uptime`
- Check IP address: `ifconfig -a`
- Pipe: `|`
 - `$ps -ef | grep sshd`
- If you are interested in programming in Linux
 - take CPS 3525 in the spring semester
 - Linux system, Unix commands
 - Languages: PHP + API, Python + API, refresh Java, C++

PHP, web, database and Linux

- References

- <http://imc.kean.edu/>
- <http://mally.stanford.edu/~sr/computing/basic-unix.html>
- <https://www.w3schools.com/html>
- <https://www.w3schools.com/php/>
- https://www.w3schools.com/php/php_mysql_intro.asp

- Do a lot of exercise

- Tools – putty (Windows), terminal (Mac), ssh, FileZilla
- Linux command
- HTML, PHP, PHP MySQL

- Please work on the project and go to see Samurai if you need any help.

Conclusion

- Familiar with Linux system and commands
- Practice Linux commands
- Study PHP and HTML
 - <https://www.w3schools.com/html/>
 - <https://www.w3schools.com/php/default.asp>
- Familiar with basic Unix commands
- Start working on Project